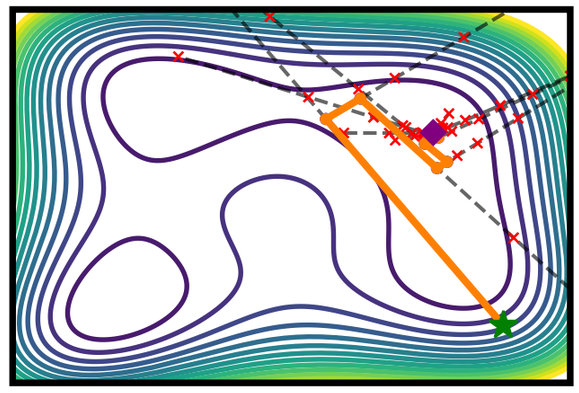
$$[\, V_i^{(k+1)},\ \theta_i^{(k+1)},\ m_i^{(k+1)} \,] = x_i^{(k+1)}$$

Backtracking Line Search



$K\times$

Expanded View:
Kth Linear Block

Δv Update Proposal

Δθ Update Proposal

Layer Norm

Tanh Activation

Δm Update Proposal

Linear

Linear

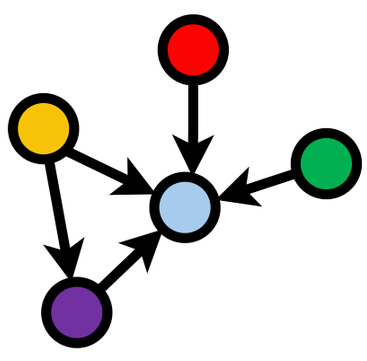
$$\mathrm{ctx}_i^{(k)} = W^{(k)}\,\mathrm{ctx}_i^{(k)} + b^{(k)}$$

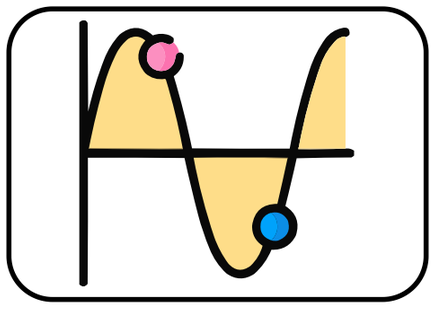
$$k = 1, 2, \ldots, K$$

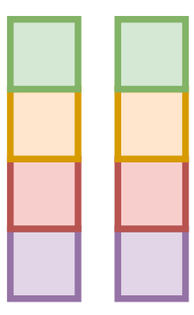
Linear

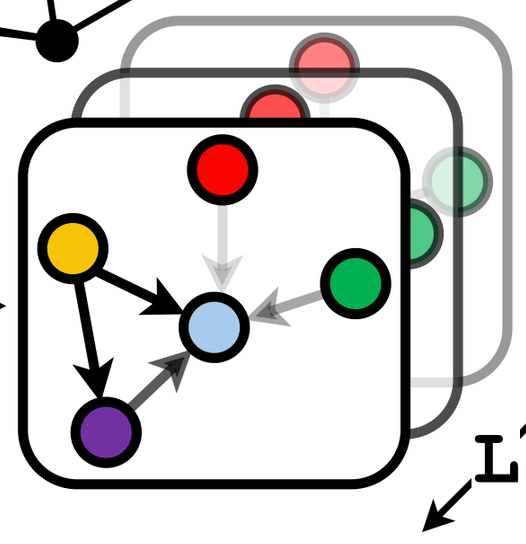
Linear

Physics Informed









L

Power Grid as Graph Input

$$\ell_{ij} = \left(G_{ij},\ B_{ij},\ B_{ij}^{sh}\right)$$

$$x_i^{(k)} = [\, m_i^{(k)},\ V_i^{(k)},\ \theta_i^{(k)},\ \ell_{ij} \,]$$

$$\Delta P_i^{(k)},\ \Delta Q_i^{(k)}$$

Message Aggr. (Self Attention)